

mmWave IC Design Final Project Report

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COLUMBIA ENGINEERING

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of Engineering and Applied Science

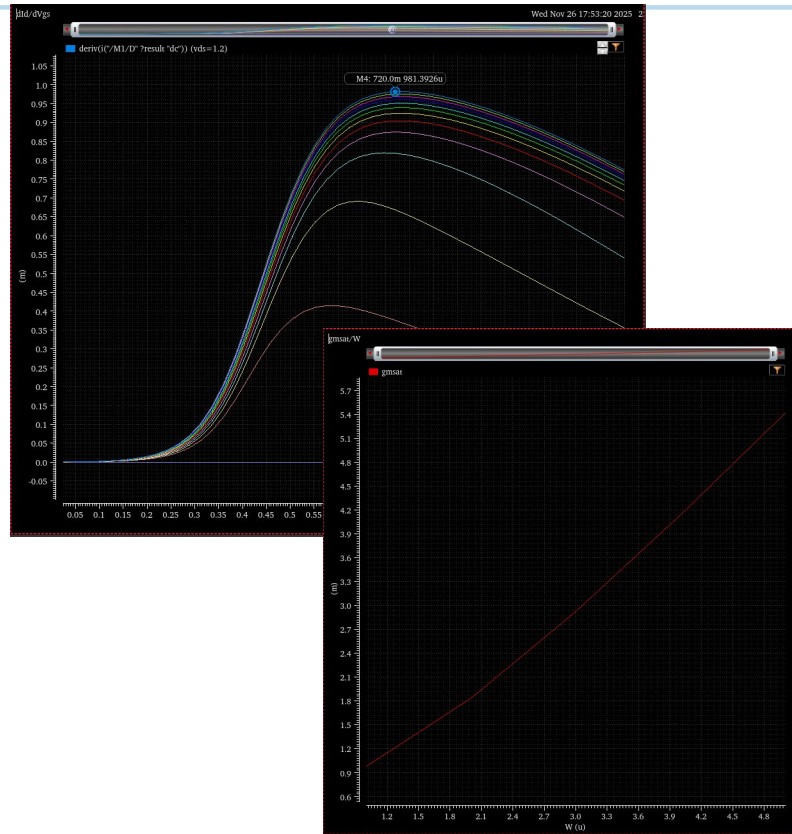


Technology Characterization



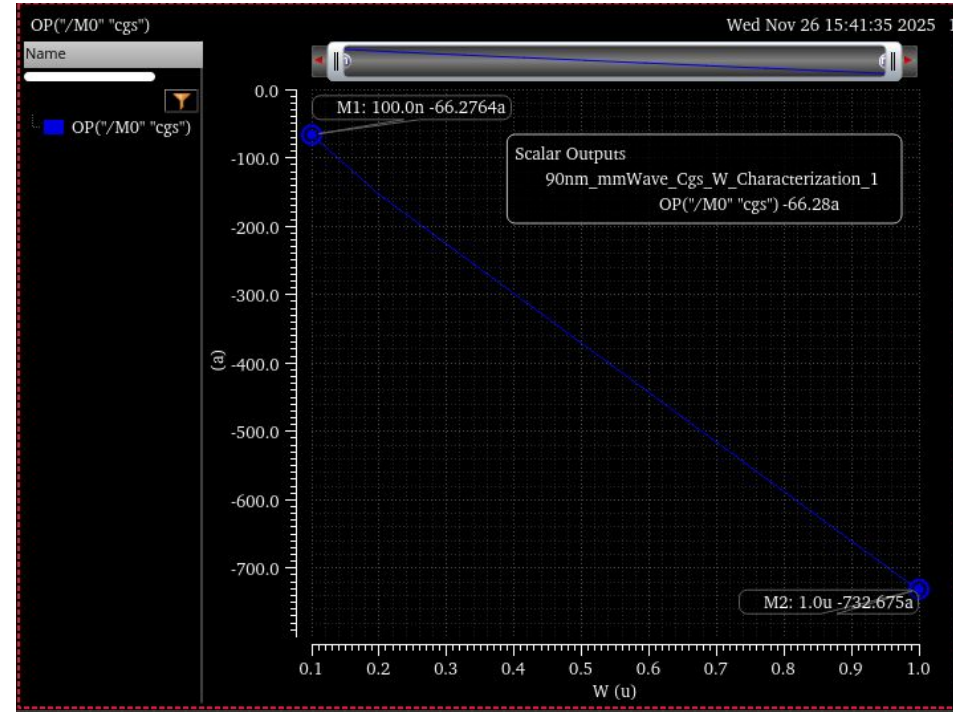
Gm sat characterization - Kaden Du

To characterize the $g_{m_{sat}}$ of the device, the dI_D/dV_{GS} was plotted over V_{GS} for multiple values of V_{DS} . As shown in the plot below, as V_{DS} approaches higher values, the gm reaches a maximum values. This maximum was taken to be the $g_{m_{sat}}$. These maximum values were plotted across a sweep of widths from 1u to 5um, and the slope of this line was taken to be the $g_{m_{sat}}/W$. For a 90nm device, the result was 981uS, resulting in a $g_{m_{sat}}/W$ value of 1.124mS/um. For a 60nm device, a similar value of 1.08mS/um was achieved. These values are somewhat close, so we can approximate an all-L value of **~1.1mS/um**.



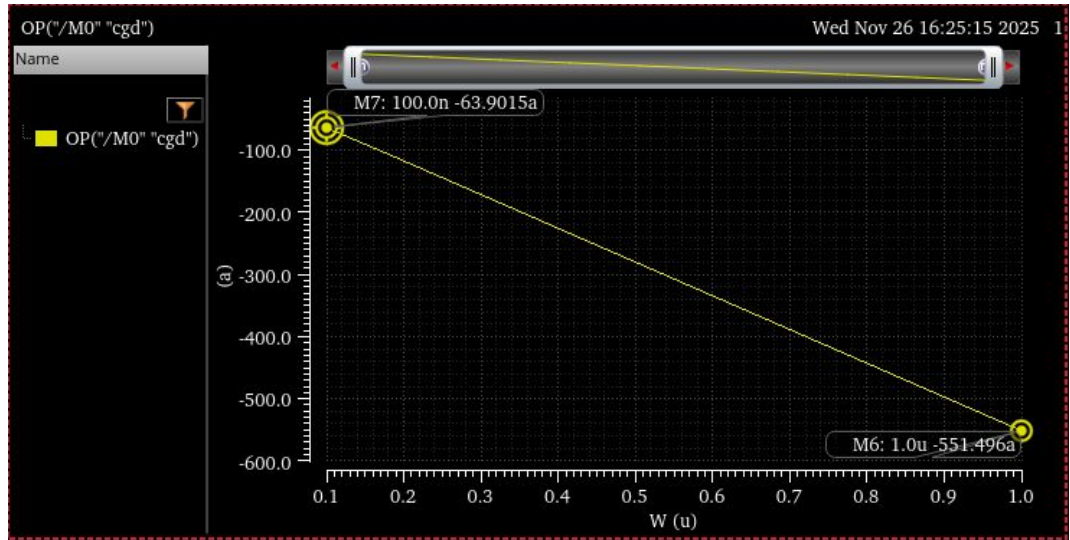
Cgs Characterization - Kaden Du

The C_{gs}/W parameter was obtained by parametrically sweeping the width of a MOSFET biased in the strong inversion region with $V_{GS} = 1.2$. The remainder of the terminals were grounded. The C_{gs} was plotted across W and the slope was taken as the result. The result was **740af/um** at 90nm length. When repeated at 60nm, the result was **345af/um**.



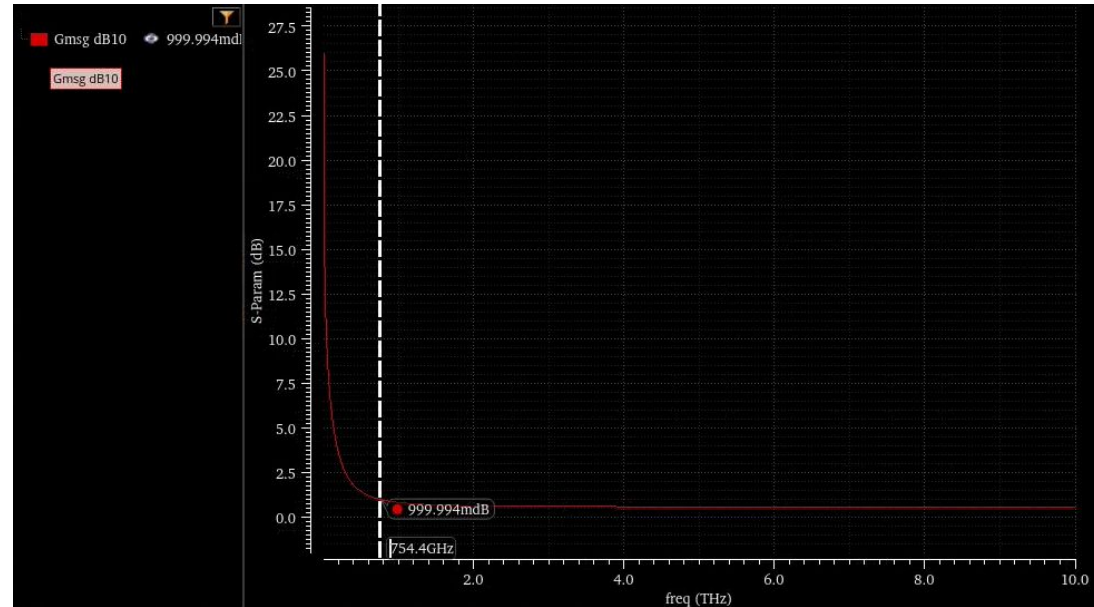
Cgd Characterization - Kaden Du

This test was run the same way as Cgs. For a 90nm device, the C_{gd}/W is **541af/um**. For 60nm, the C_{gd}/W is **332af/um**.



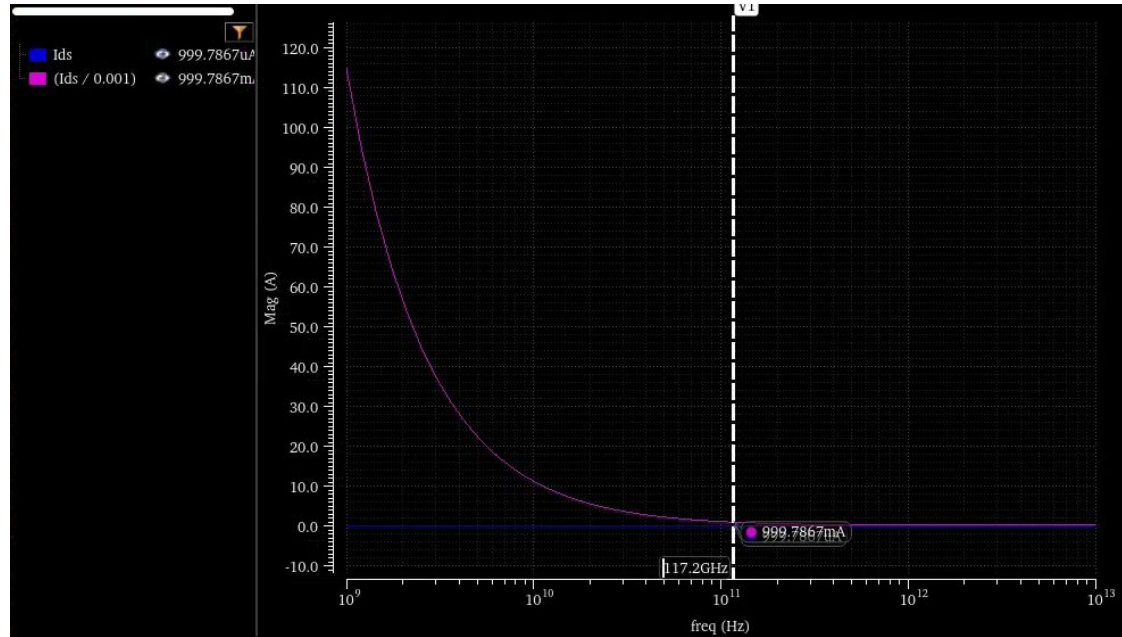
fMax Characterization - James Jagielski

The Fmax is 754.4GHz



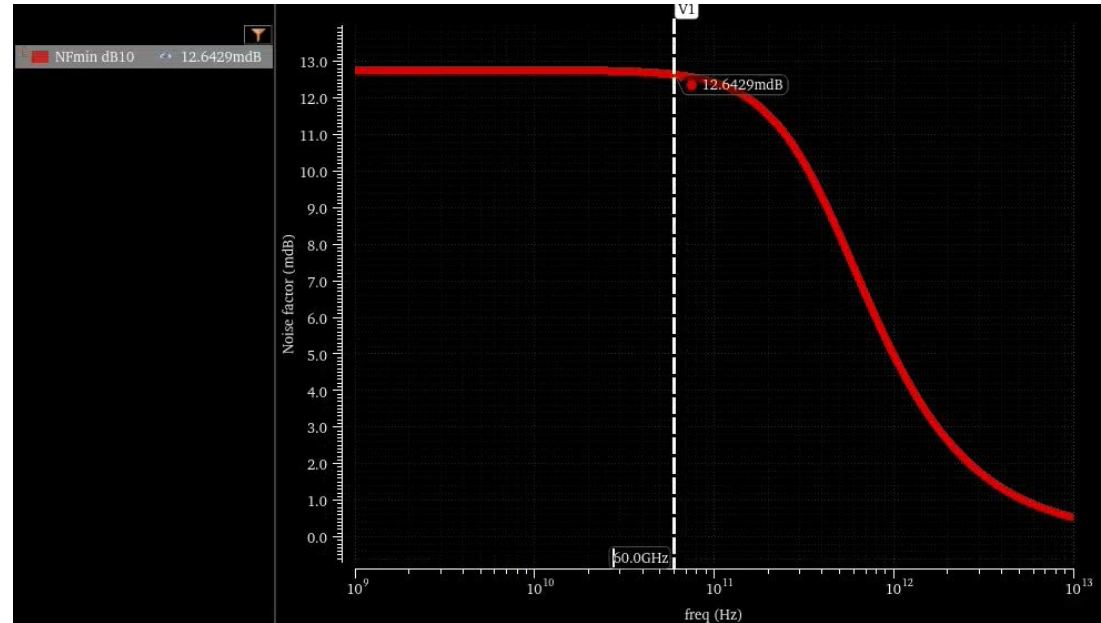
fT Characterization - James Jagielski

The f_t is equal to 117.2GHz



NF min Characterization - James Jagielski

The NF min is equal to 12.6429mdB at 60GHz



γ Characterization - Stijn Heyde

NMOS

Current noise $A^2/Hz = 5.27 \cdot 10^{-24}$

$G_m = 0.861 mS$

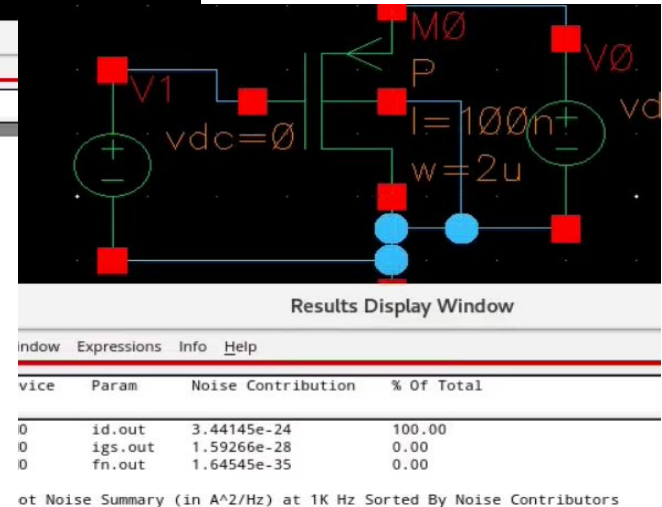
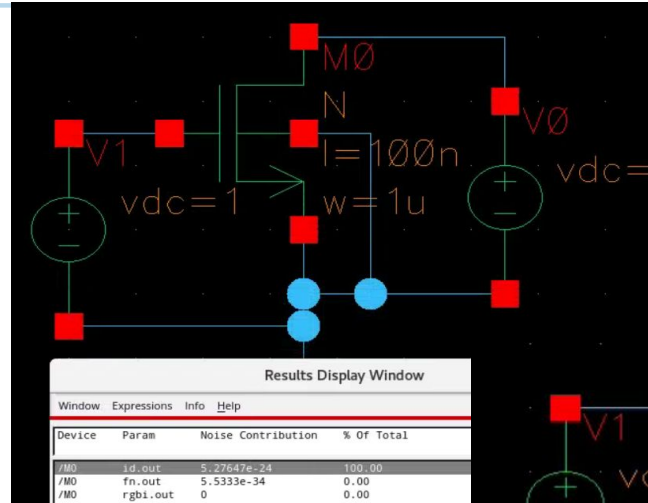
$$\gamma = \frac{i_{n,d}^2}{4kTRg_m} = 0.37$$

PMOS

Current noise $A^2/Hz = 3.44 \cdot 10^{-24}$

$G_m = 1.378 mS$

$$\gamma = \frac{i_{n,d}^2}{4kTRg_m} = 0.15$$

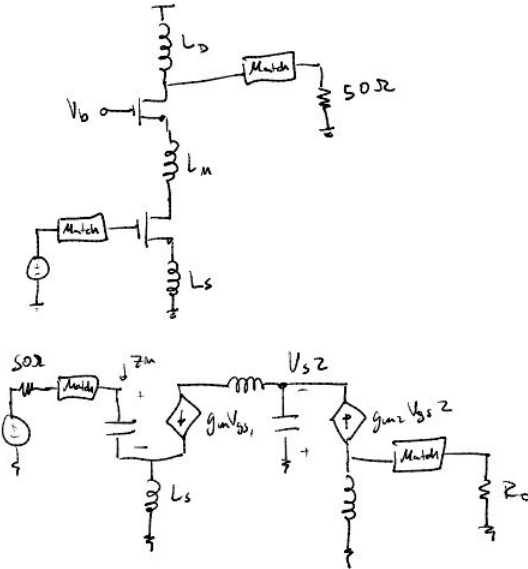


Circuit Design



Theoretical Analysis - LNA - Kaden Du

Theoretical Analysis



Matching Condition

$$Z_{in} = \frac{V_{in}}{I_{in}} = \frac{V_{gs1} (g_m V_{gs1} + s C_{gs} V_{gs1}) s L_s}{s C_{gs} V_{gs1}}$$

$$= \frac{1}{s C_{gs}} + \frac{g_m L_s}{C_{gs}} + s L_s$$

$$\boxed{\frac{g_m L_s}{C_{gs}} = 50 \Omega}$$

Gain

$$V_s = \frac{V_{in}}{2} \quad (\text{by matching assumption})$$

$$V_s = V_{gs1} (g_m + s C_{gs}) s L_s$$

$$V_s = \left(\frac{V_{in}}{2} - V_s \right) (g_m + s C_{gs}) s L_s$$

$$V_s (1 + (g_m + s C_{gs}) s L_s) = \frac{V_{in}}{2} (g_m + s C_{gs}) s L_s$$

$$V_s = \frac{\frac{V_{in}}{2} (g_m + s C_{gs}) s L_s}{1 + (g_m + s C_{gs}) s L_s}$$

$$V_{gs1} = \frac{V_{in}}{2} \left(1 - \frac{(g_m + s C_{gs}) s L_s}{1 + (g_m + s C_{gs}) s L_s} \right)$$

$Z_{in} \approx \frac{V_{in}}{2}$ when $s L_s, g_m, s C_{gs}$ are small

$$Z_{in} = s L_m + \left(\frac{1}{g_m} \parallel \frac{1}{s C_{gs2}} \right)$$

$$\frac{1}{g_m + s C_{gs}} = \frac{g_m - s C_{gs}}{g_m^2 + \omega^2 C_{gs}^2}$$

$$Z_{in} = s L_m + \frac{g_m - s C_{gs}}{g_m^2 + \omega^2 C_{gs}^2}$$

Design such that inductor resonates capacitance out

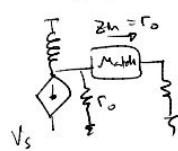
$$Z_{in} = \frac{g_m}{g_m^2 + \omega^2 C_{gs}^2} \quad V_s \approx \frac{g_m}{2} \frac{V_{in}}{Z} \left(\frac{1}{g_m^2 + \omega^2 C_{gs}^2} \right)$$

→ 20

Theoretical Analysis - LNA - Kaden Du

$$V_s = \frac{g_{m1}}{g_{m2}} \frac{V_{in}}{Z}$$

$$I = -g_{m2}(V_{s2})$$



Assuming matched, half power to output

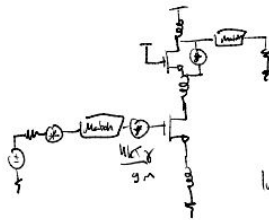
$$V_{out} = -g_{m2} \left(\frac{V_{s2}}{Z} \right) r_o$$

$$V_{out} \approx -g_{m1} \frac{r_o}{4} V_{in}$$

where r_o is cascode impedance with inductor

$$R_o \approx (r_{o1} + r_{o2} + g_{m2}(r_{o1}, r_{o2})) \parallel L_s$$

Noise Analysis



output noise M_z

$$I = 4kT \gamma g_m^2$$

output noise voltage

$$V^2 = 4kT \gamma g_m \left(\frac{R_o}{Z} \right)^2$$

$$\text{input referred} = \frac{V^2}{G^2}$$

$$\frac{16kT \gamma g_{m2}}{g_{m1}^2} \leftarrow \frac{16kT \gamma g_m R_o^2}{g_{m1}^2 R_o^2}$$

$$NF = 1 + \frac{\gamma}{g_{m1} Z_o} + \frac{4 \gamma g_{m2}}{g_{m1}^2 Z_o}$$

Takeaways

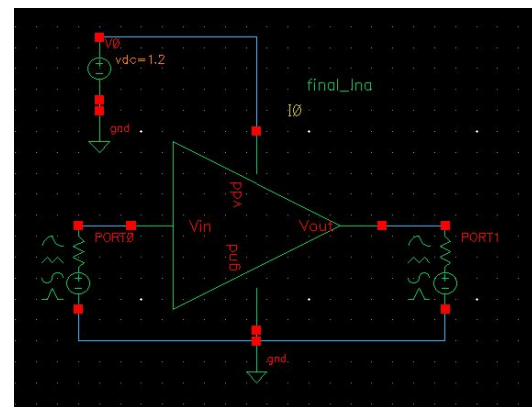
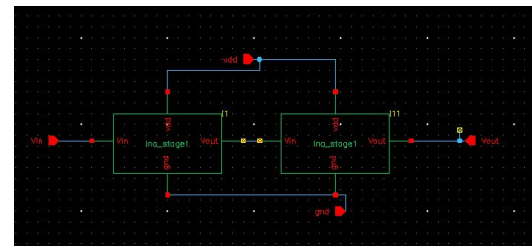
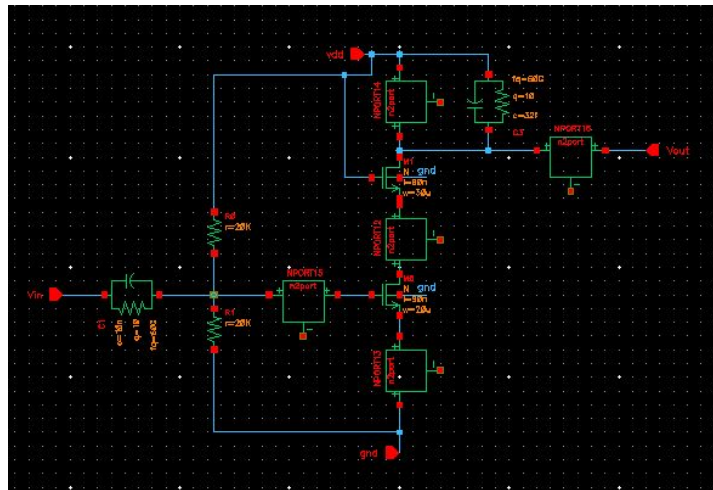
- Design g_{m1} , L_s , C_{gs1} to give 50Ω input impedance for good noise and power match
- Design L_m to resonate C_{gs2} to give a real input impedance of common gate stage (possible Q boosting if $\frac{1}{g_m} \gg \gg \gg$)
- Design R_o (g_{m2} , r_{o1} , r_{o2}) to give a matchable output and high gain.



Simulation Results - LNA - Kaden Du

Schematics and Testbench

- Used Spiral Inductors with touchstone file in n-port block for L
- Cascaded design to improve gain with a small NF tradeoff
- 18.12mA total current consumption
- Design from [1]

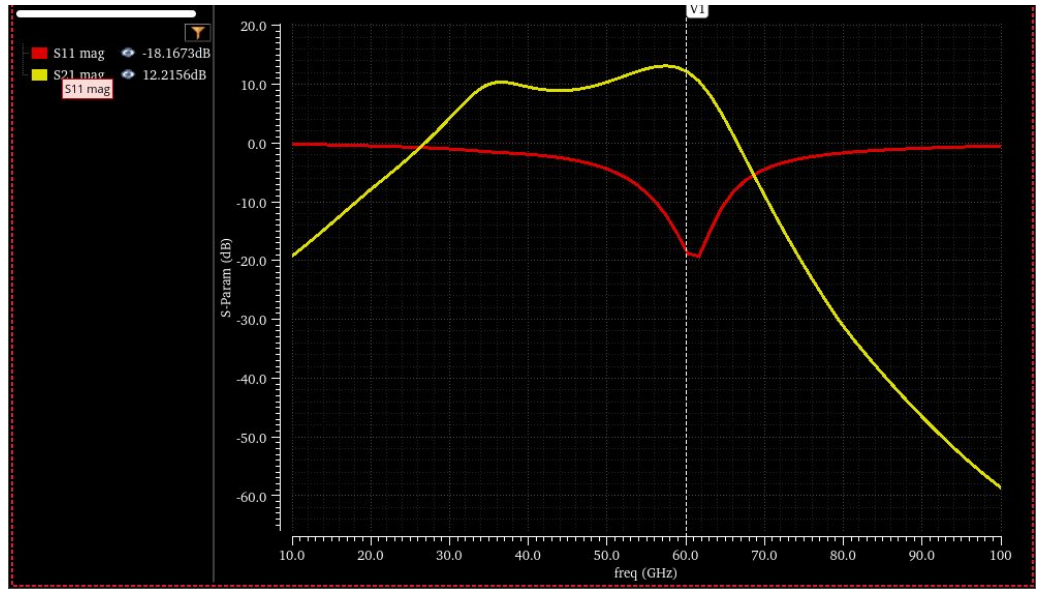


Simulation Results - LNA - Kaden Du

S parameters

- Input and Output both shown to be matched to $\sim 50\Omega$
- **-18.16dB S11**, proving matching
- Power gain of **12.21dB (S21)**

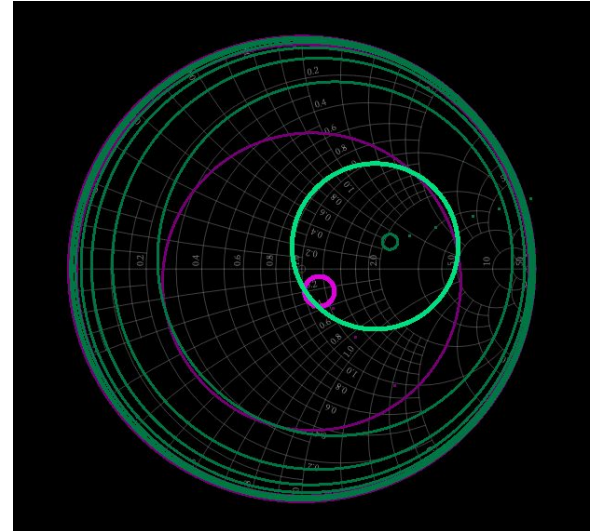
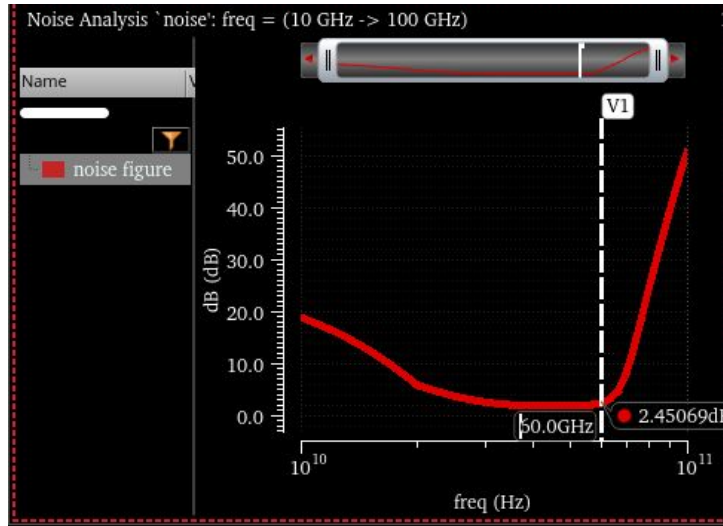
ZM 1 reOhms	expr	value(real(zm(1 ?result "sp"))) 6e...	58.79
ZM 1 imOhms	expr	value(imag(zm(1 ?result "sp"))) 6e...	-10.1
ZM 2 reOhms	expr	value(real(zm(2 ?result "sp"))) 6e...	56.94
ZM 2 imOhms	expr	value(imag(zm(2 ?result "sp"))) 6e...	10.16



Simulation Results - LNA - Kaden Du

Noise Analysis

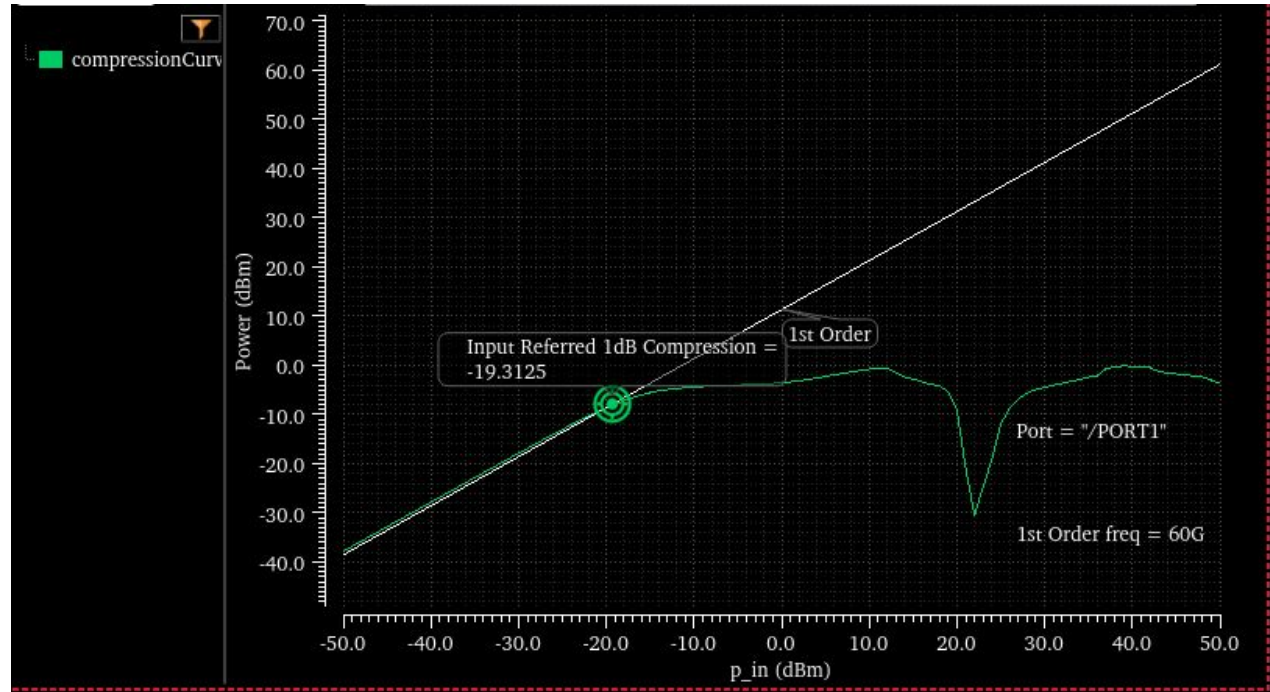
- **2.45dB NF**
- Source inductor allows for good noise/power match
- 50 Ohm input impedance inside 2.5dB NF circle, close to 12.5dB gain, so perfect input match was chosen



Simulation Results - LNA - Kaden Du

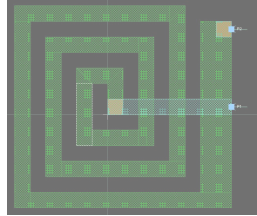
-1dB Compression Point

- -1dB compression point at **-19.31dB**, well within spec of >-30dB

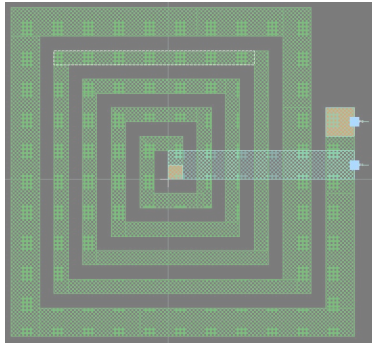


Spiral Inductor Design - LNA - Kaden Du

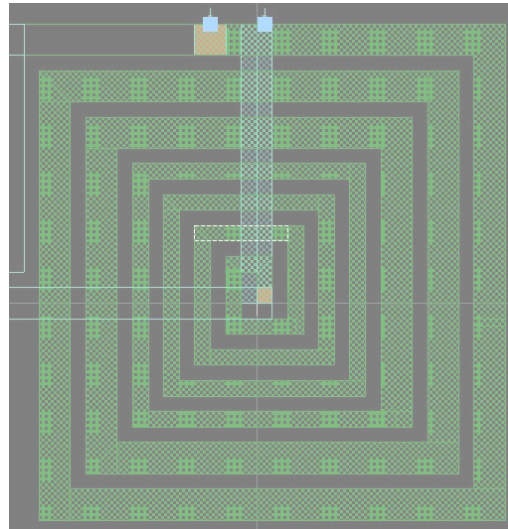
59pH Inductor (Input Match) - 13umx13um



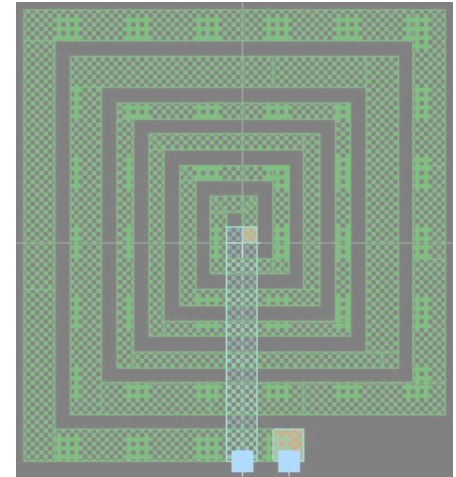
195pH Inductor (Top and Bottom) - 24umx24um



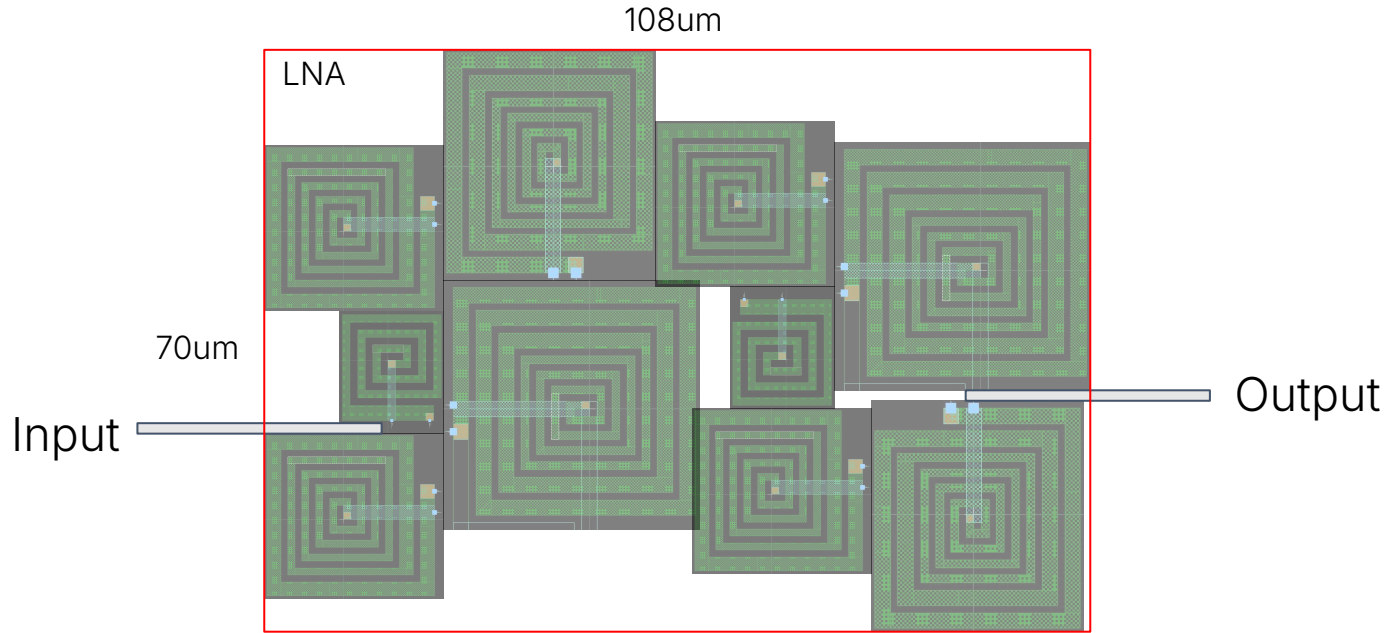
450pH Inductor (Middle) - 32umx32um



300pH inductor (output match) - 28umx28um



LNA Floorplan- Kaden Du



Theoretical Analysis - Mixer - Stijn Heyde

- RF transconductor: $g_{m,RF}=14 \text{ mS}$
- LO switching pair: $g_{m,LO}=12 \text{ mS}$
- Load resistors: $R_L=300 \text{ } \Omega$

Ideal Case (Hard Switching LO)

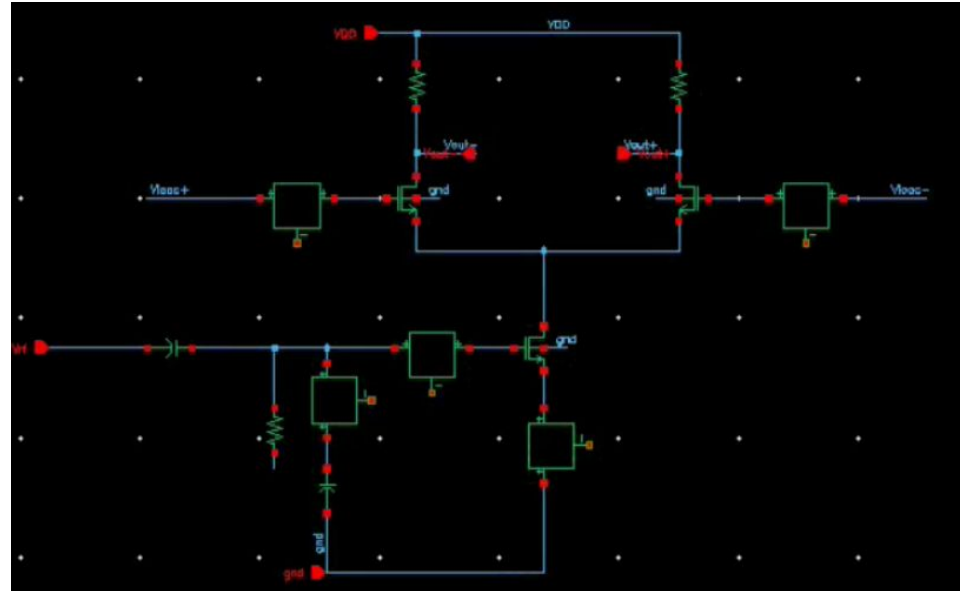
$$A_v \approx 2\pi / g_{m,rf} \cdot R_L \approx 2\pi \cdot 0.014 \cdot 300 \approx 2.67 \text{ (8.5 dB)}$$

⇒ will probably be a lot lower since hard switching is not realistic

Simulation Results - Mixer - Stijn Heyde

Schematics

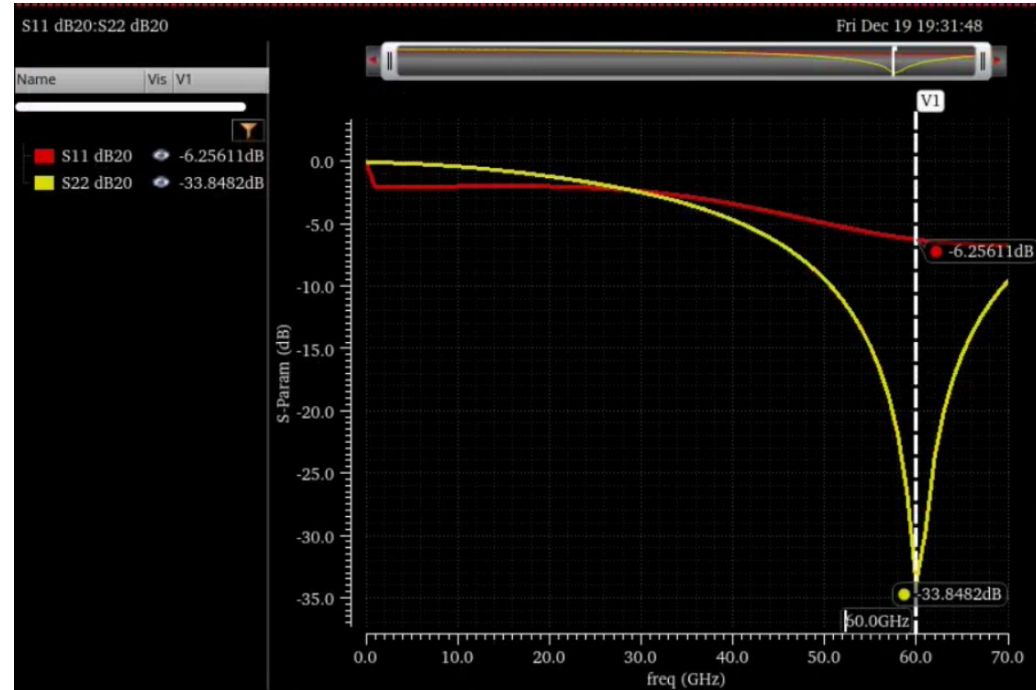
- Used Spiral Inductors with touchstone file in n-port block for L



Simulation Results - Mixer - Stijn Heyde

S-parameters

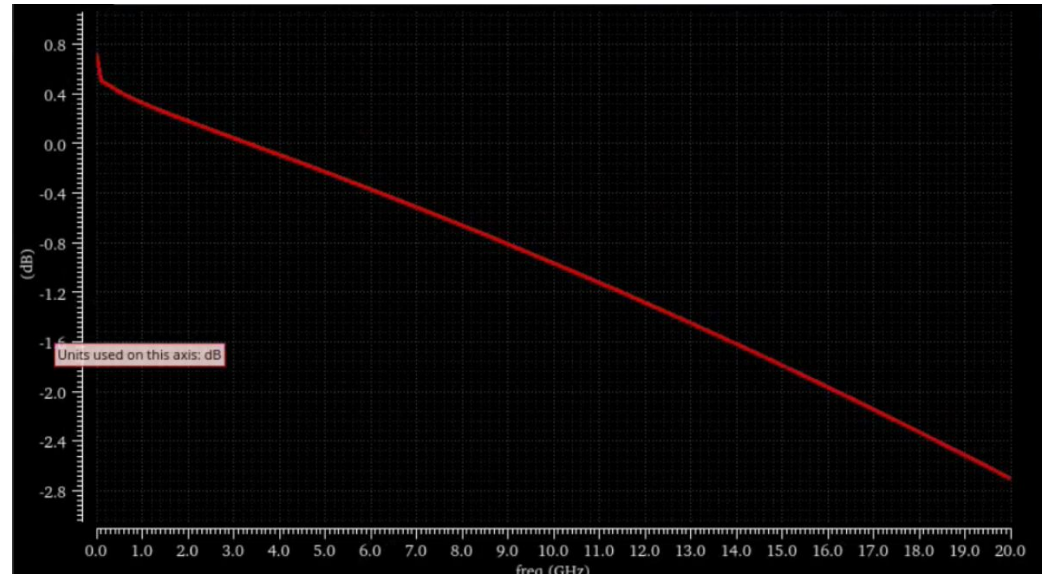
- S11 RF-port matched to 50 Ohm
-6.26dB around 60 GHz
- S22 LO-port matched to 100 Ohm differentially
-34dB at 60 GHz



Simulation Results - Mixer - Stijn Heyde

Conversion-Gain

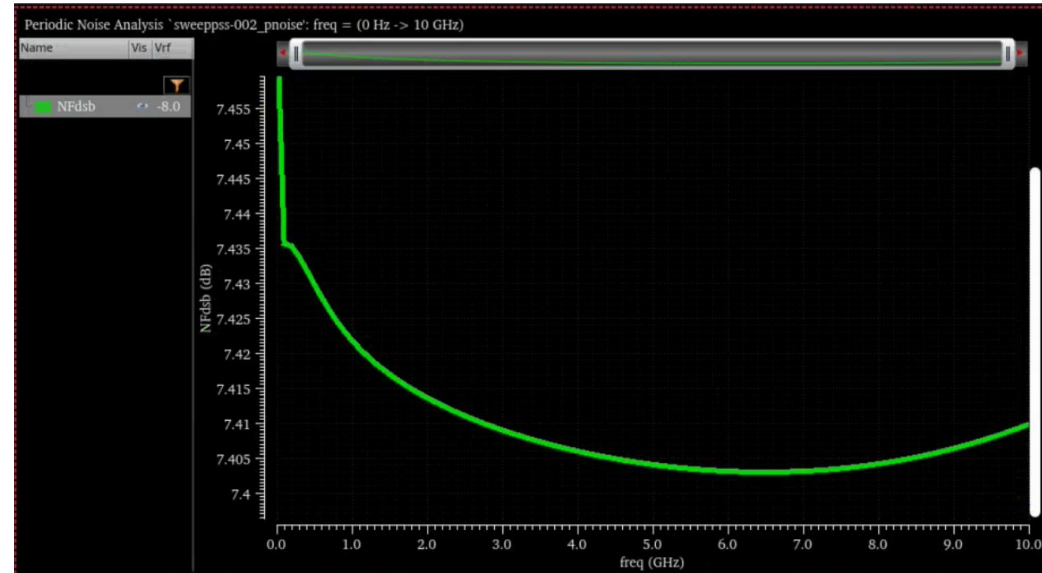
- Maximum gain of 0.4dB



Simulation Results - Mixer - Stijn Heyde

Double Sideband Noise-Figure

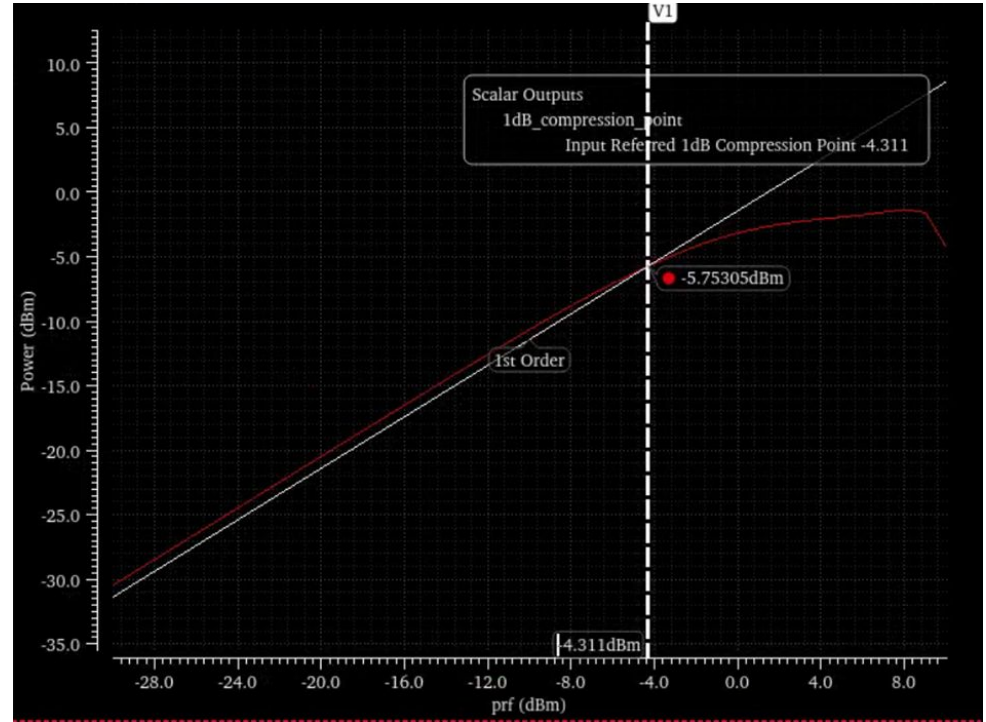
- 7.4dB double sideband NF



Simulation Results - Mixer - Stijn Heyde

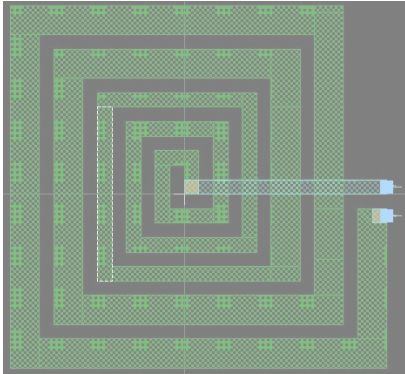
-1dB Compression Point

- -5.8dBm

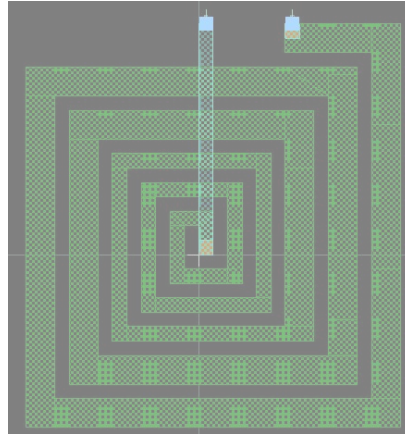


Inductors Results - Mixer - Stijn Heyde

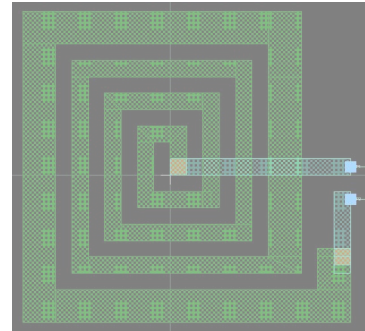
200pH, 26ux26uH



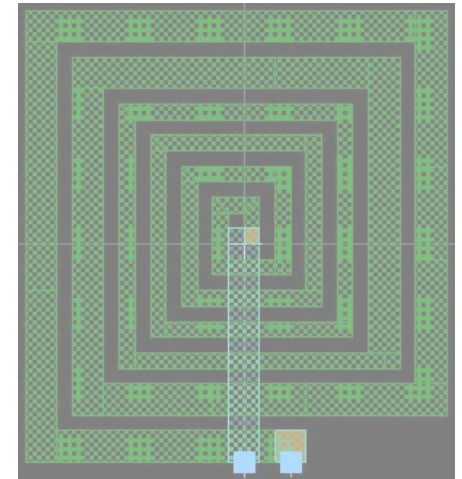
220pH, 27ux26uH



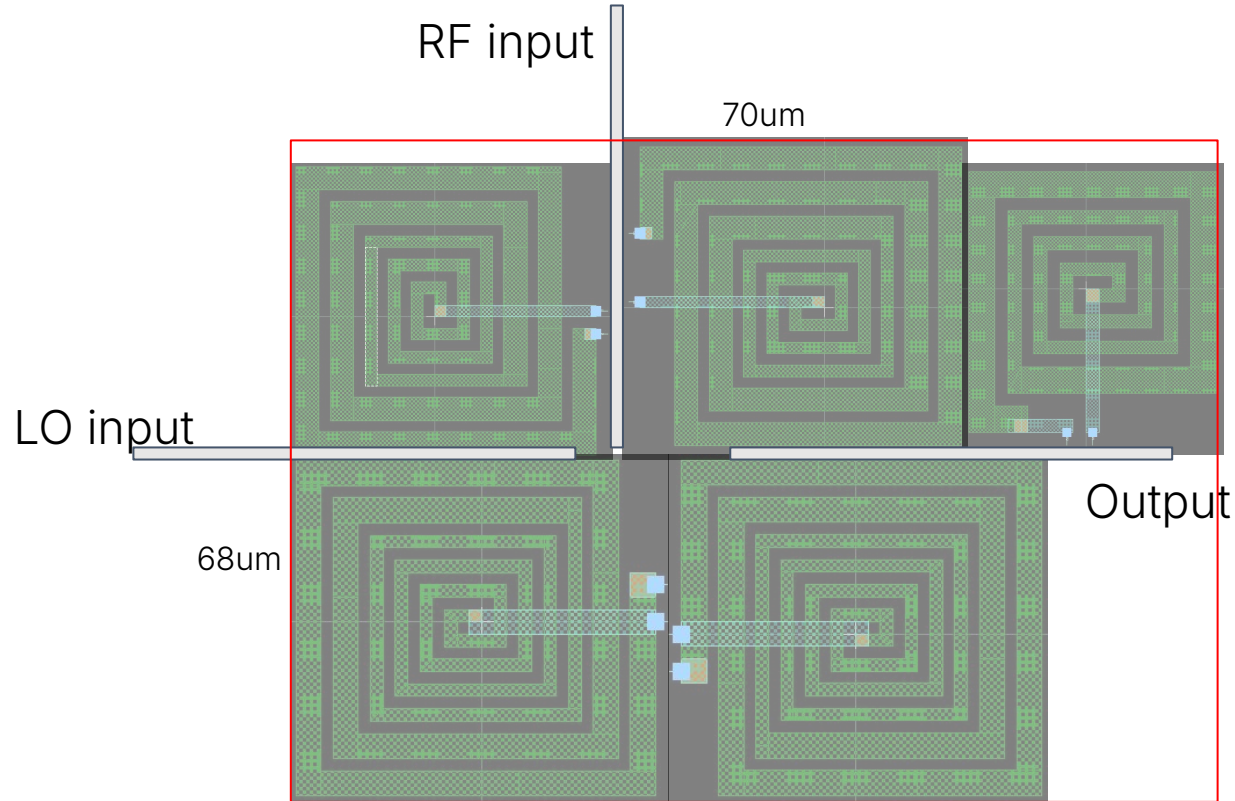
100pH, 19ux19uH



300pH inductor (output match) - 28umx28um



Floorplan - Mixer - Stijn Heyde



Theoretical Analysis - VCO - James Jagielski

Target 60 GHz

$L = 40 \text{ pH}$

$Q = 15$

$$C = \frac{1}{(2\pi f_0)^2 L} \Rightarrow \frac{1}{(3.77 \times 10^{11})^2 \times 40 \times 10^{-12}}$$

$$C = 180 \text{ fF}$$

$$R_s = \frac{\omega L}{Q}$$

$$R_s = 1 \Omega$$

$$R_s = \frac{3.77 \times 10^{11} \times 40 \times 10^{-12}}{15}$$

$$R_p = Q^2 R_s$$

$$R_p = 225 \Omega$$

$$R_{eq} \geq R_p$$

$$R_{eq} = \frac{2}{g_m}$$

$$g_m \geq \frac{2}{225} = 8.9 \text{ mS}$$

Phase noise

$$P = \frac{V^2}{R} = \frac{0.29^2}{225} = 0.35 \text{ mW}$$

$$P = \frac{0.57^2}{225} = 1.44 \text{ mW}$$

$$\frac{2.5 \times 4 \times 10^{-21}}{3.5 \times 10^{-3}} = 2.9 \times 10^{-17}$$

$$\frac{2.5 \times 4 \times 10^{-21}}{1.44 \times 10^{-3}} = 6.9 \times 10^{-18}$$

$$\left(\frac{f_0}{2Q\Delta f} \right)^2 = \left(\frac{60 \times 10^9}{2 \times 15 \times 10^6} \right)^2 = 2000^2 = 4 \times 10^6$$

$$2.9 \times 10^{-17} \times 2 \times 4 \times 10^6 = 2.3 \times 10^{-10}$$

$$\mathcal{L}(1 \text{ MHz}) = 10 \log_{10}(2.3 \times 10^{-10}) = -96 \text{ dBc/Hz}$$

$$6.9 \times 10^{-18} \times 2 \times 4 \times 10^6 = 5.5 \times 10^{-11}$$

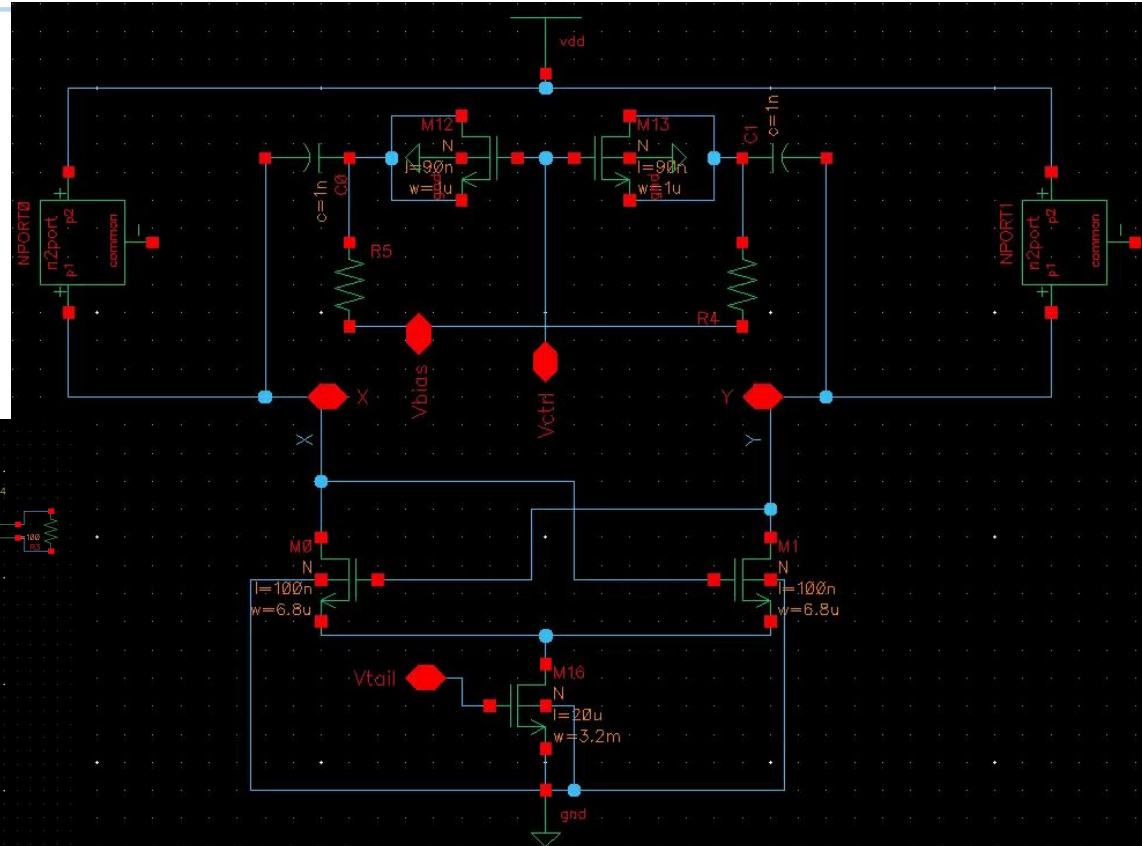
$$\mathcal{L}(1 \text{ MHz}) = 10 \log_{10}(5.5 \times 10^{-11}) = -102.6 \text{ dBc/Hz}$$



Design - VCO - James Jagielski

We used a simple Cross Coupled VCO design (pictured on the right) using varactors for variable capacitors gaining tunable range. We were able to achieve all specifications and output to 60GHz. We have a current draw of 3.821mA producing a power of 4.5mW.

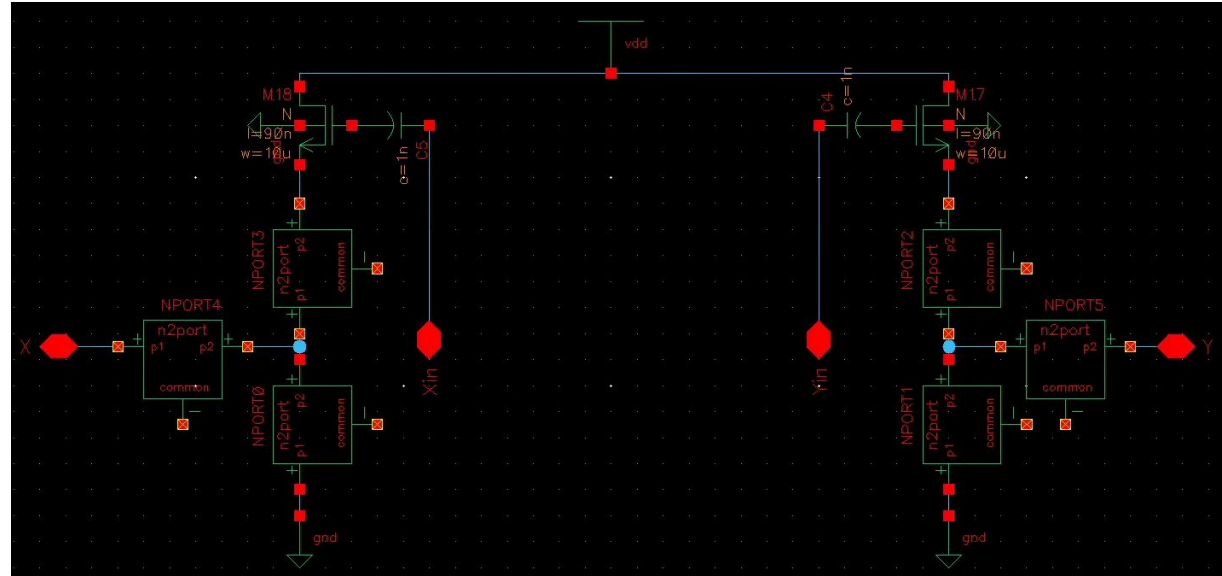
Below we have the full test set up with the buffers included.



Design - First Buffer - James Jagielski

This buffer was used to avoid loading the VCO to maintain higher frequencies and phase noise. This created some attenuation of the signal, but we used another buffer to get the signal back to usable amplitudes.

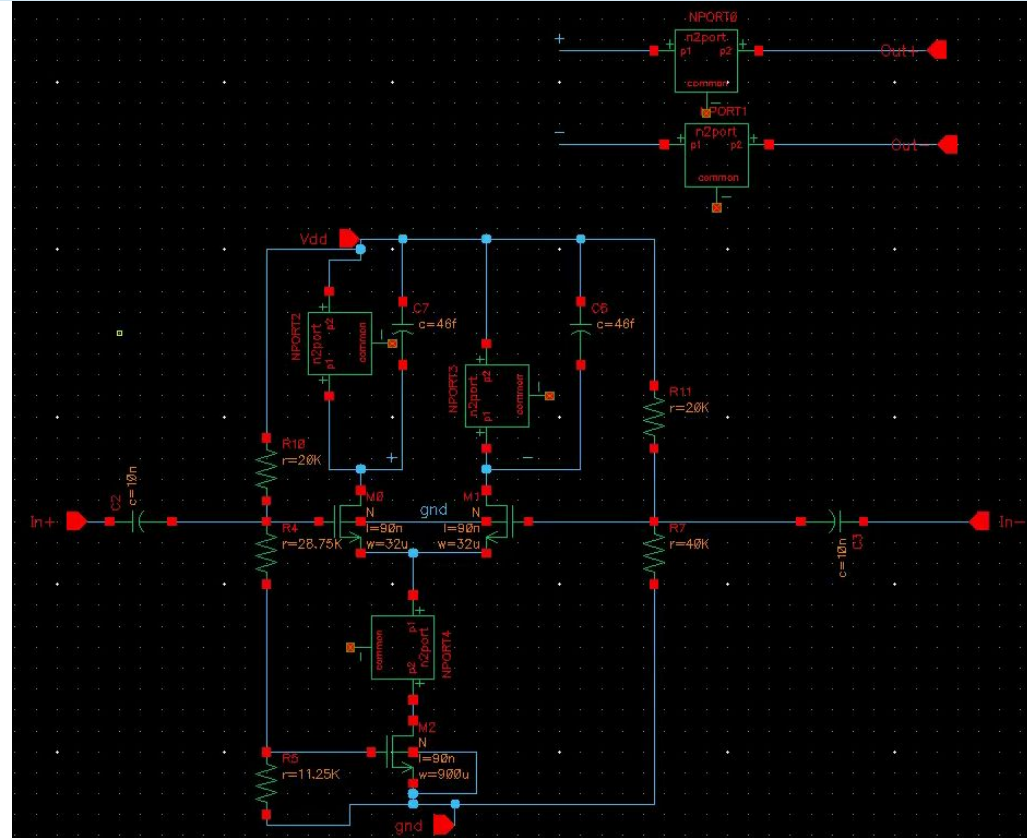
Power: 1mW



Design - Second Buffer - James Jagielski

This second buffer was used to provide gain to the signal, so the mixer could actually be able to use it. This is a high gain stage, which would've loaded the VCO with the gate capacitance.

Power: 10mA



Tuning Range Results - VCO & Buffer - James Jagielski

- **Absolute tuning range**

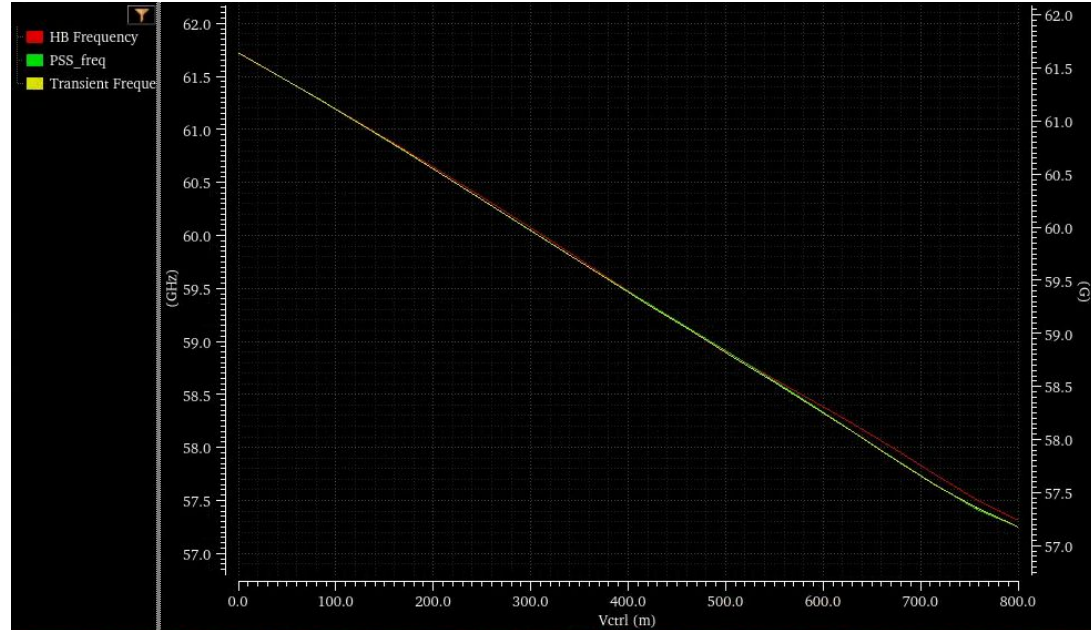
- $\Delta f = f_{\max} - f_{\min} = 61.6 - 57.25 = 4.35 \text{ GHz}$

- **Center frequency:**

- $f_{\text{center}} = (f_{\max} + f_{\min}) / 2 = (61.6 + 57.25) / 2 = 59.425 \text{ GHz}$

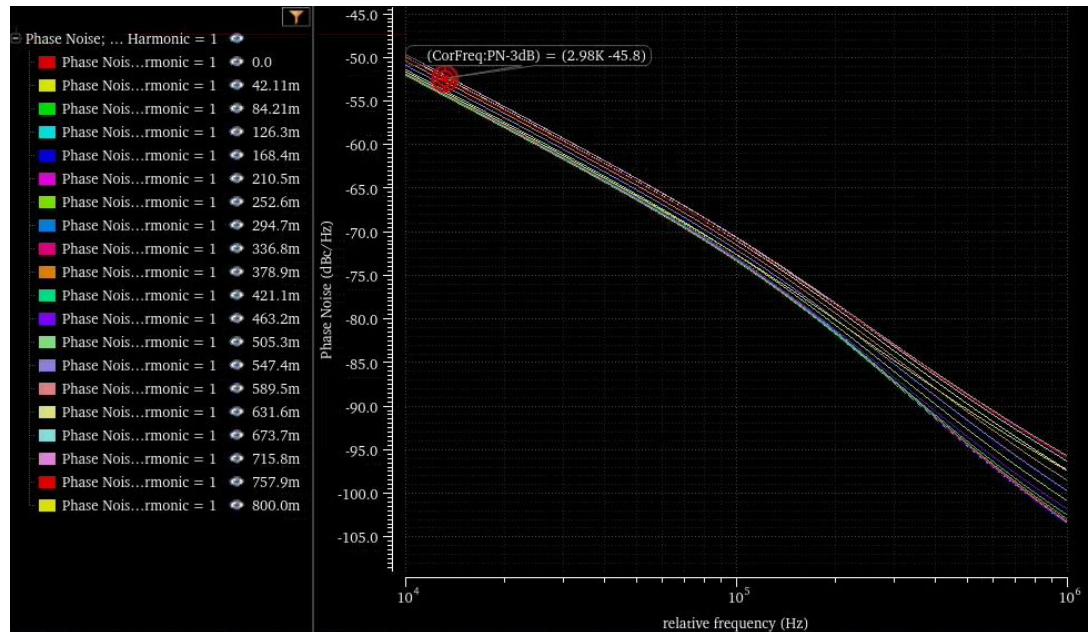
- **Fractional range:**

- $\text{FTR} = 4.35 / 59.425 \times 100\% \sim 7.32\%$



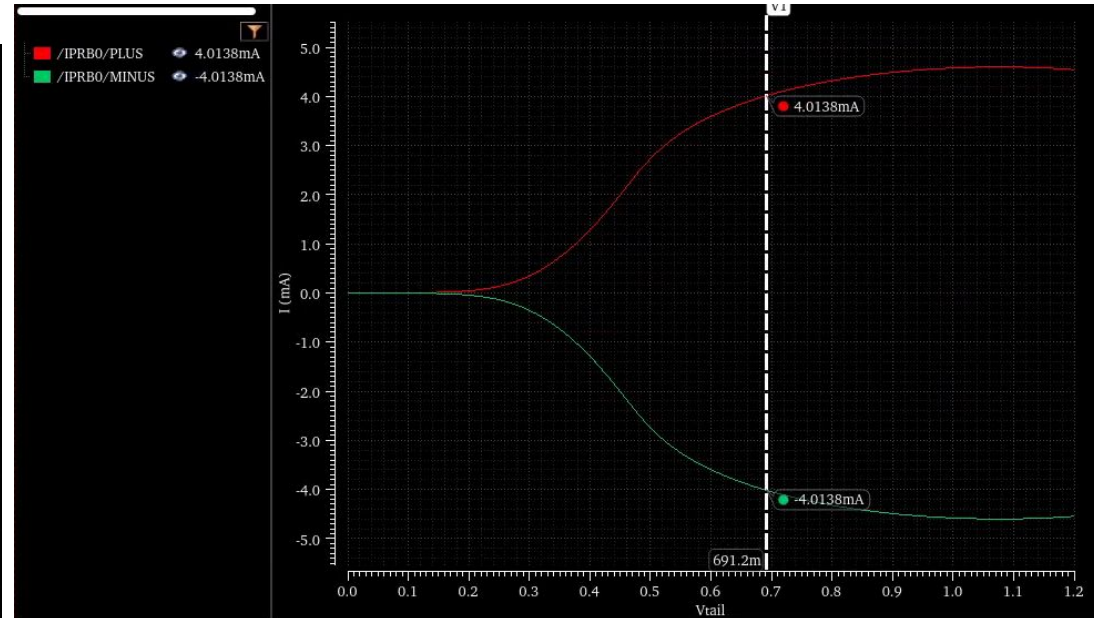
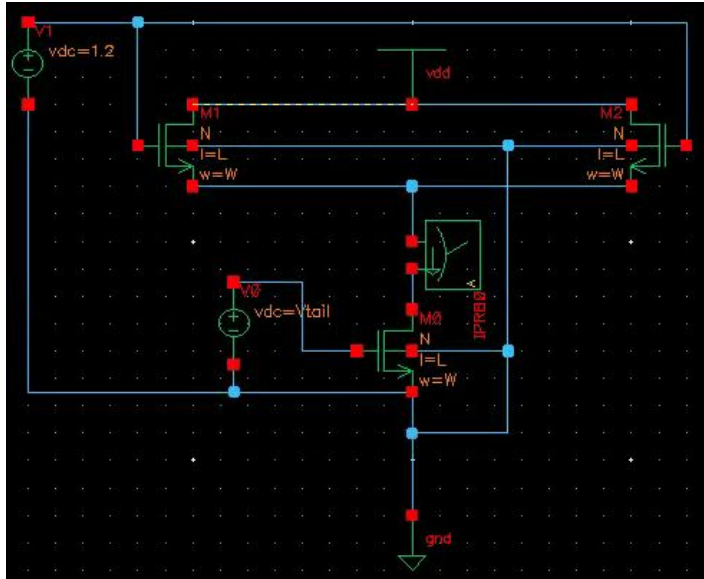
Phase Noise Results - VCO & Buffer - James Jagielski

This plot demonstrates the VCO well beneath the phase noise criteria of -90dB/Hz. The maximum within the tuning range getting around -94dB/Hz.



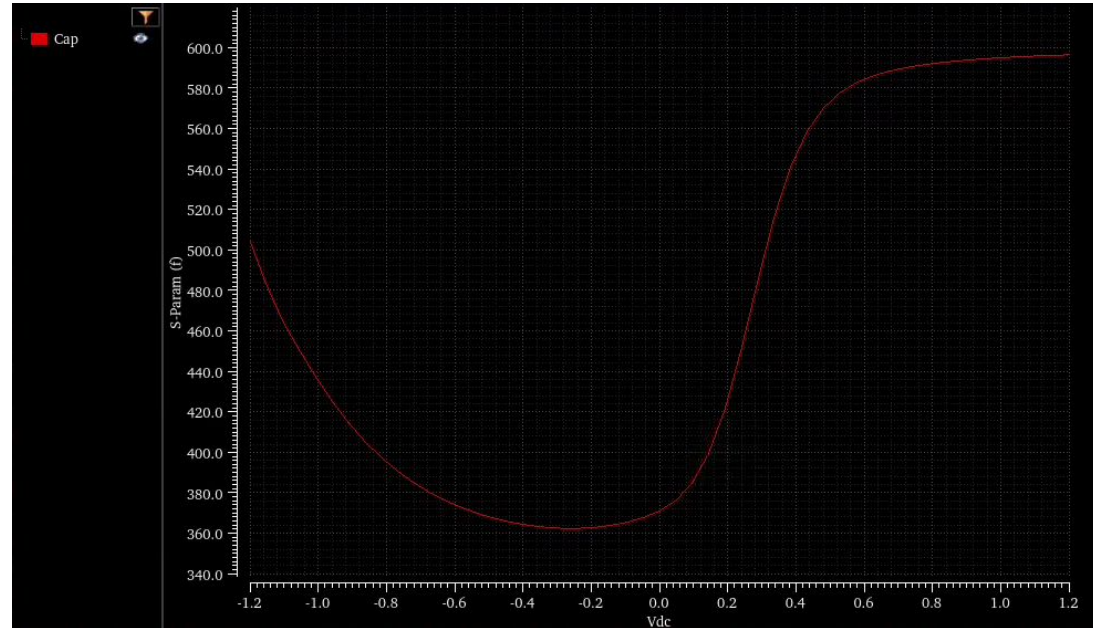
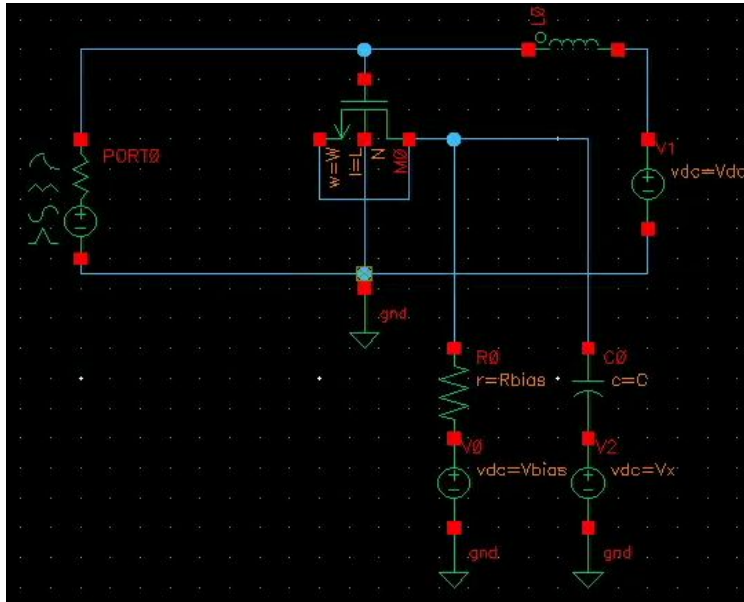
Vtail Results - VCO - James Jagielski

We simulated our bias current to get the correct sizing of the transistor. We achieved $\sim 4\text{mA}$ at a bias of 691.2mV for the bias transistor in the differential pair. We achieved $\sim 3.8\text{mA}$ when integrated with the whole system.



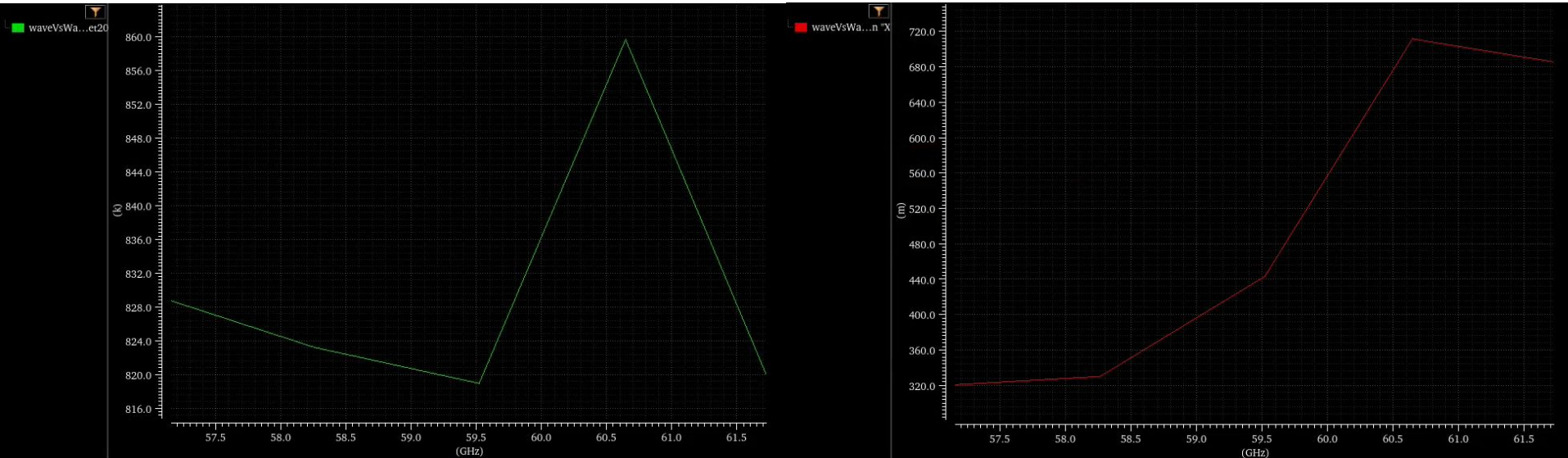
Varactor - VCO - James Jagielski

We simulated our varactor isolating it to get to test the behavior and what control voltage range we could use. We used 0V to 800mV for our tuning range. This ranges from 360fF to 580fF.



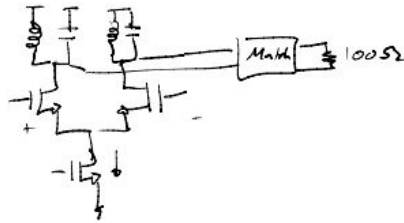
Amplitude vs Frequency - VCO - James Jagielski

We simulated the amplitude of the output of the second buffer stage and the VCO output.

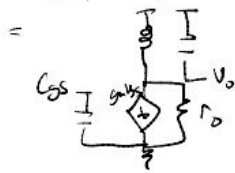


Theoretical Analysis - LO Buffer - Kaden Du

Theoretical Analysis - Buffer



Half circuit



Design L_D , C_L to resonate.

$$V_o = 2 g_m r_o V_{in}$$

$$A_v = 2 g_m r_o$$

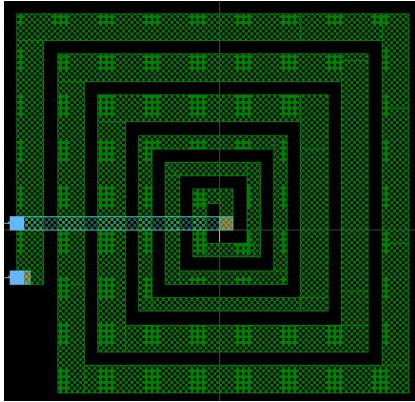
Notes

- Design r_o to be 100 Ω , reduce matching effort.
- Design W_L to be low such that C does not load heavily. (no input match)
- High bias current for high gain.

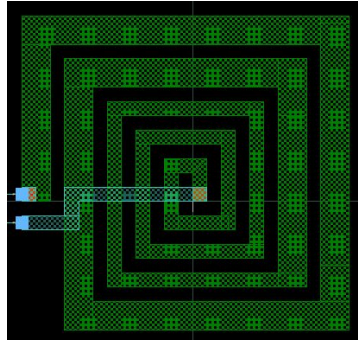


VCO and LO Buffer Floorplan

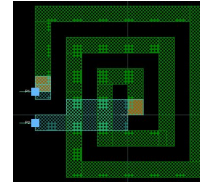
273pH inductor - 29umx29um



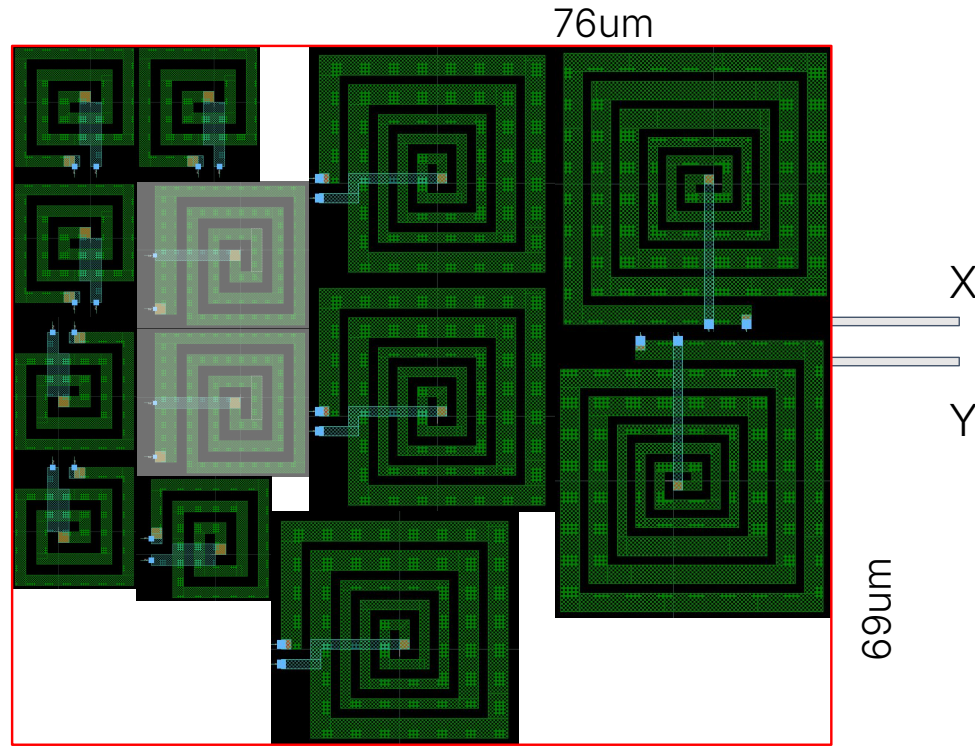
150pH inductor - 23umx23um



40pH inductor - 11umx11um



VCO and LO Buffer Floorplan



Theoretical Analysis - Baseband Amplifier - Stijn Heyde

3 stages

- 1st gain stage

$$A_{v,1} \approx g_m \cdot r_{out} = 24\text{dB}$$

- 2nd gain stage

$$A_{v,2} \approx g_m \cdot r_{out} = 28\text{dB}$$

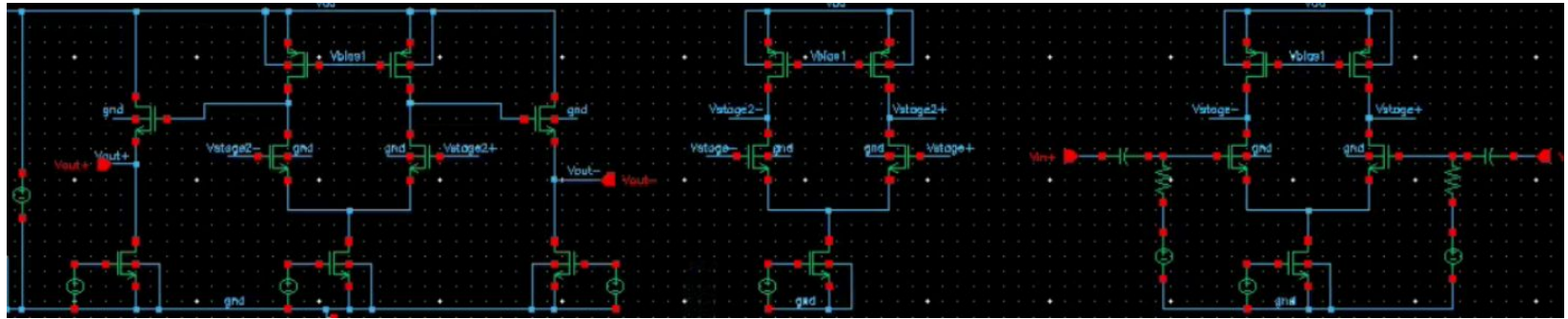
- Source follower for low 50 Ohm load (100 Ohm differential)

$$A_{v,drv} \approx g_m (r_{out} \parallel 50\Omega) \text{ dominated by } 50\Omega \Rightarrow A_{v,drv} = 0\text{dB}$$

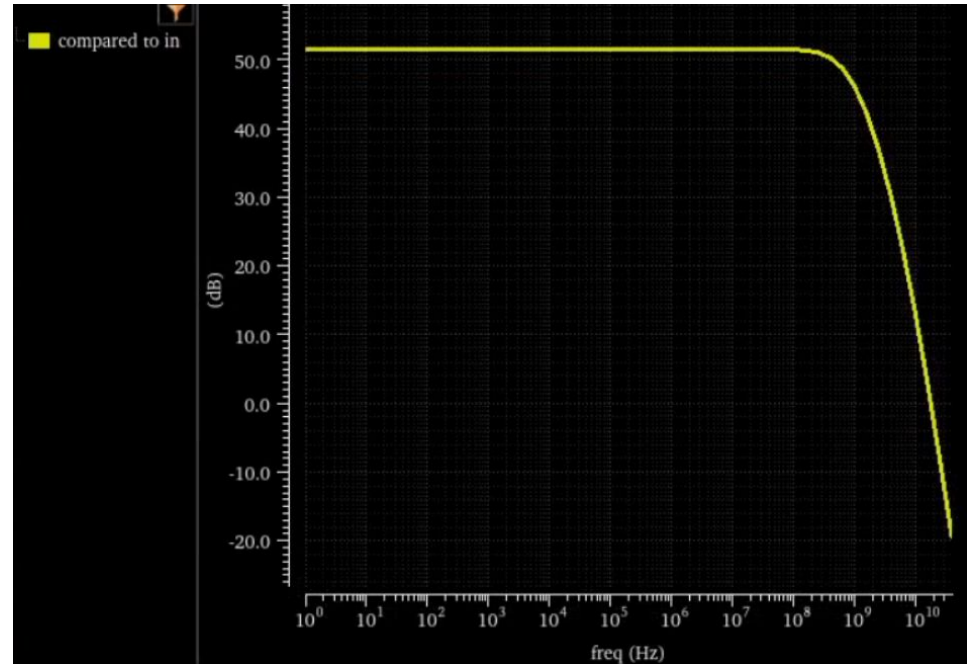
First stage dominates noise

Simulation Results - Baseband Amplifier - Stijn Heyde

Schematic



Simulation Results - Baseband Amplifier - Stijn Heyde

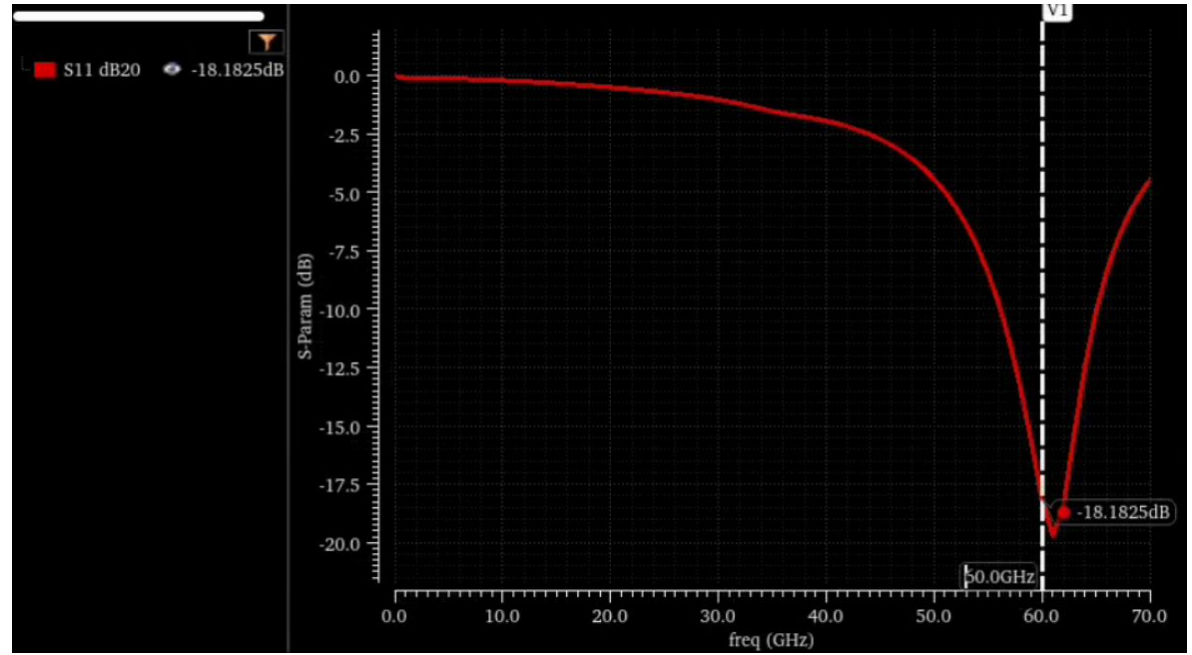


Overall Receiver Simulation



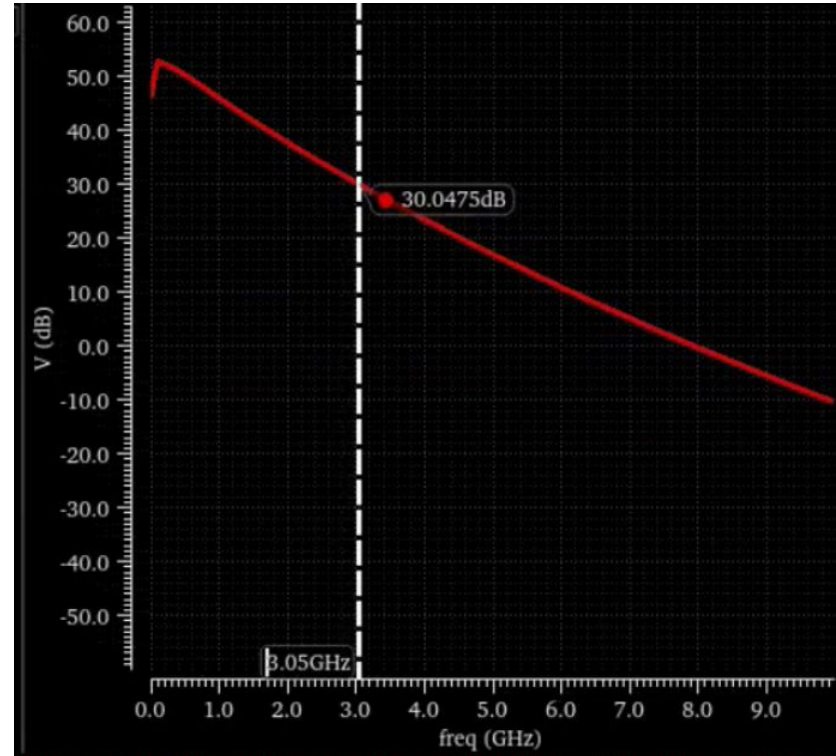
Overall Receiver - Matching

S11 LNA input port matched
to 50 Ohm at -18.2dB



Overall Receiver - Conversion Gain

The 30dB conversion gain is achieved up to a 3GHz bandwidth



Overall Receiver - NF

Theoretical Noise Figure Calculation

Friis Equation: $NF = NF_1 + (NF_2 - 1)/G_1 + (NF_3 - 1)/G_1G_2$

Using measured values for NF and Gain of LNA, Mixer, and Baseband Amplifier

- $NF_{LNA} = 2.45\text{dB}$, $\text{Gain}_{LNA} = 12.21\text{dB}$
- $NF_{Mixer} = 7.4\text{dB}$, $\text{Conversion Gain}_{Mixer} = 0.6\text{dB}$
- $NF_{BBA} = 11.3\text{dB}$

Total Noise figure = 4.28 dB

Noise figure simulation for downconversion depends on desired bandwidth, and much could be filtered out. As bandwidth was not specified, we could not integrate the noise power over said bandwidth.

Overall Receiver - IP1dB

The design achieves a 1dB compression point up to -29.5dBm



Overall Receiver - Power

LNA: 22mW

Mixer: 7mW

Baseband amplifier: 9mW

VCO = 4.5mW

First buffer = 1mW

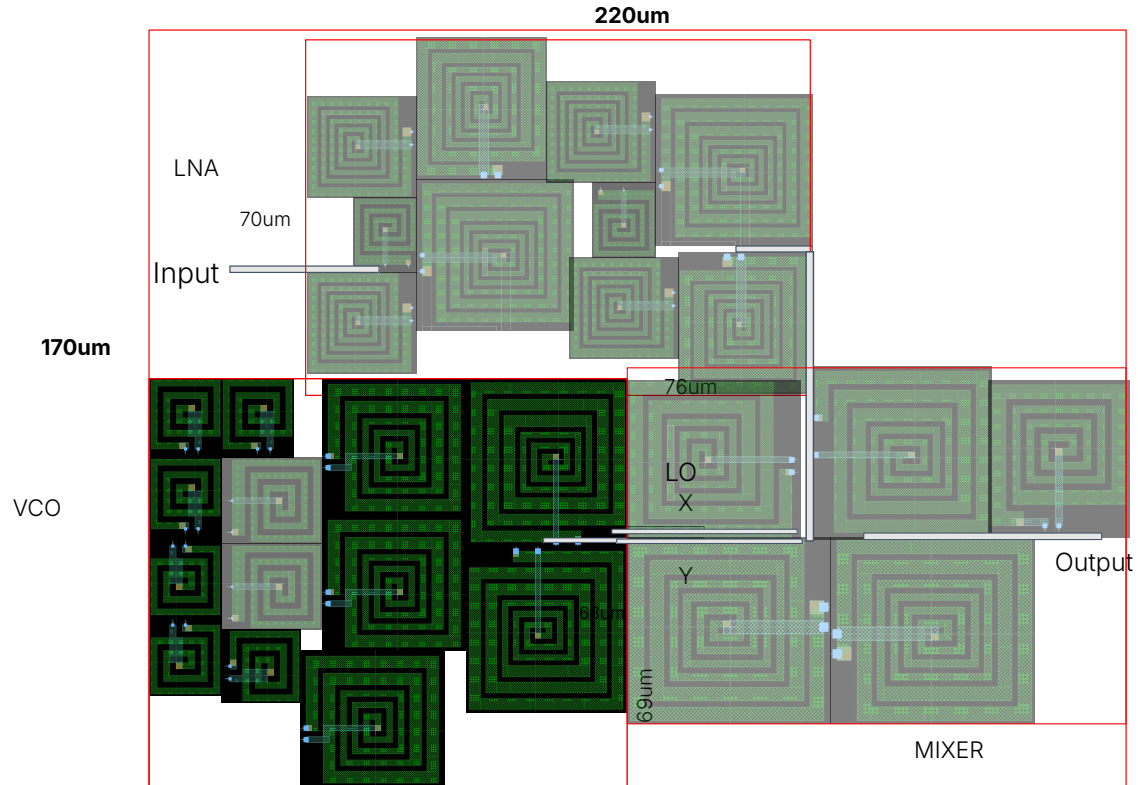
Second buffer 10mW

⇒ **total power consumption: 53.5mW**



Floorplanning

LNA



References

- [1] - T. Yao et al., "Algorithmic Design of CMOS LNAs and PAs for 60-GHz Radio," in IEEE Journal of Solid-State Circuits, vol. 42, no. 5, pp. 1044-1057, May 2007, doi: 10.1109/JSSC.2007.894325.
- [2] M. Kraemer, D. Dragomirescu, and R. Plana, "Design of a very low-power, low-cost 60 GHz receiver front-end implemented in 65 nm CMOS technology," International Journal of Microwave and Wireless Technologies, vol. 3, no. Special Issue 2, pp. 131-138, Apr. 2011, doi:10.1017/S1759078711000067.
- [3] B. Razavi, "A 60-GHz CMOS receiver front-end," IEEE Journal of Solid-State Circuits, vol. 41, no. 1, pp. 17-22, Jan. 2006.

